ENGINEERING GEOLOGY

UNIT - V : DAMS, RESERVOIRS, TUNNELS

Geology of Dams, Reservoirs and Tunnels

Syllabus (as given)

- Types and purpose of dams; geological considerations in the selection of a dam site; geological considerations for successful construction of reservoirs, life of reservoirs; purpose of tunnelling, effects, lining of tunnels; influence of geology for successful tunnelling.

1. Dams - Types and Purpose

A **dam** is a **barrier constructed across a river or valley to store water**, forming a **reservoir**. India has > 5000 large dams (3rd in the world); e.g., **Nagarjuna Sagar, Srisailam, Tungabhadra, Somasila, Bhakra, Tehri, Hirakud**.

1.1 Purposes of Dams

- **Irrigation** (storage for canals), **drinking and industrial water supply**.
- **Hydro-electric power** generation.
- **Flood control / moderation** and **river regulation**.
- **Navigation, fisheries, recreation, tourism**, ground-water recharge, **silt control**.
- **Multi-purpose dams** serve more than one aim (Nagarjuna Sagar: irrigation + power + flood control).

1.2 Types of Dams

- **Based on material: masonry / concrete** (gravity, arch, buttress), **earth-fill, rock-fill**, steel, timber.
- **Based on structural action:**

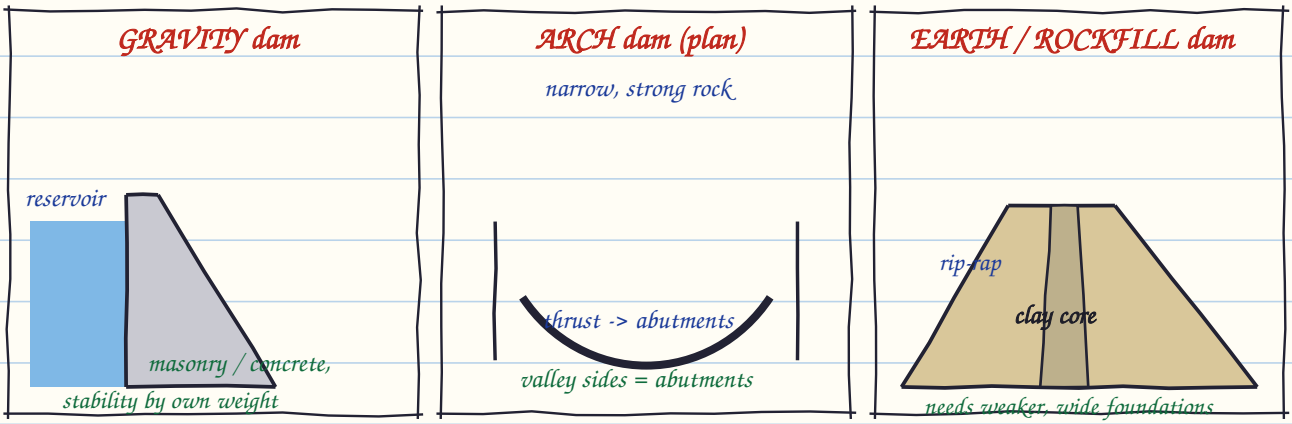


Fig 1: Gravity, arch and earth / rock-fill dams

Type	Features and suitable geology
Gravity dam	**Resists water thrust by its own weight**; masonry / concrete; triangular section; needs **strong, sound rock foundation** (granite, gneiss, quartzite, massive basalt) - very high bearing capacity. Eg: Nagarjuna Sagar, Bhakra
Arch dam	**Curved in plan**, transfers thrust to **abutments**; thin section, less concrete; needs **narrow gorge with strong, sound, unjointed rock walls**. Eg: Idukki (Kerala), Vajont
Buttress dam	Water face supported by triangular buttresses; needs **good rock at buttress points**; less concrete than gravity
Earth-fill (embankment) dam	**Compacted earth** with **impervious core** and shell; **flexible**; can be built on **weaker, deformable foundations (alluvium, soil)**; suitable in **wide valleys** and seismic zones. Eg: Somasila, Tungabhadra (partly)
Rock-fill dam	Rock dumped / compacted with an **impervious membrane or core**; use where good rock is available; Eg: Tehri (earth + rock fill)
Composite dam	Combination - masonry / concrete section + earth embankment (Nagarjuna Sagar, Tungabhadra)

- **Other classification:** by **hydraulic design** (overflow / non-overflow), by **height** (low < 30 m, medium 30-90 m, high > 90 m).

Foundation rule

- Concrete gravity / arch dams -> hard, strong rock. Earth / rock-fill -> any competent foundation incl. soil. The **higher the dam, the better the foundation must be**.

2. Geological Considerations in Selection of a Dam Site

A good dam site must satisfy **geological, topographical, hydrological and economic** conditions. The geologist studies **rock type, structure, strength, permeability and weathering** of the foundation and abutments.

2.1 Topographic Conditions

- **Narrow valley (gorge)** with **wide reservoir basin** upstream (economical length, large storage).
- Strong, **sound rock at shallow depth** in river bed and abutments; **good approach** and availability of **construction materials** (aggregate, clay, sand) nearby.

2.2 Geological Considerations

1. **Rock type & strength**
   - Foundation rock must have **adequate compressive strength, low compressibility** and bearing capacity > dam load: **granite, gneiss, quartzite, dolerite, massive basalt** are excellent.
   - **Poor rocks:** shale, schist, slate, phyllite, weathered / soft sandstone, friable rocks - **slake, swell, slide**; **limestone** - cavities; **gypsum, rock salt** - soluble, unsuitable.
2. **Geological structures**
   - **Attitude of beds (strike & dip):** beds **dipping upstream** are favourable (water pressure and thrust press the dam into rock; leakage resisted); **dip downstream** -> **seepage, uplift, sliding** risk; **horizontal beds** - safe if strong; **steeply dipping beds with strike parallel to the river** - leakage.
   - **Folds:** avoid dam in **anticline crest**; prefer **syncline limb** with moderate dip; unfavourable fold may cause **leakage and instability**.
   - **Faults: avoid active / major faults** - weak crushed zone, **seepage, differential settlement, seismic displacement**. Minor faults can be treated (concrete plug, grouting).

- **Joints & fractures:** closely spaced, open, persistent joints cause leakage and low strength - treat by curtain grouting, consolidation grouting; joint sets dipping towards valley reduce abutment stability.
- **Unconformities, dykes, shear zones:** weak planes; dykes may stop or allow seepage - study.

3. **Weathering:** deep weathered rock - excavation to fresh rock (increases cost); thickness of overburden / alluvium must be small; bore holes.
4. **Permeability & ground water:** high permeability foundation -> seepage, piping, uplift pressure; ground-water level and water tightness of reservoir rim. Karst limestone areas often leak.
5. **Abutment stability:** slopes should be stable with no landslide mass (Vajont - study of reservoir rim); check dip of beds towards valley.
6. **Seismicity:** area should be tectonically stable; design for seismic coefficient; avoid active faults and reservoir-induced seismicity zones.
7. **Construction materials:** availability of suitable aggregates, sand, clay, rock nearby; quarry geology.
8. **Reservoir basin:** water-tight, with no leakage and limited siltation (see next section).

Rock	For concrete dam	Remarks
Granite / Gneiss	Excellent	Sound; grout joints
Basalt (massive)	Very good	Check vesicular / jointed layers
Quartzite / Dolerite	Excellent	Hard; may be highly jointed
Sandstone	Good if hard	Check cement; bedding planes
Limestone	Doubtful	Cavities, solution channels
Shale / Schist / Slate	Poor	Sliding, swelling; weak planes

2.3 Foundation Treatment

- **Excavation** to sound rock; **cut-off trench** (earth dams); **grout curtain** (cement grouting in a line of holes) to **reduce seepage**; **consolidation grouting** to strengthen jointed rock; **drainage holes** to reduce **uplift**; **dental concrete** in cavities; **shear keys**, rock bolts / anchors.
- **Dam failures** studied earlier (St. Francis, Malpasset, Teton, Vajont) show how neglect of geology leads to disaster.

2.4 Geological Causes of Dam Failures

- **Foundation failure:** shear along weak beds (shale / clay), **excess settlement** on soft strata, **sliding** on bedding or joint planes dipping downstream.
- **Seepage and piping:** flow through **joints, faults, solution cavities, permeable layers** -> removal of fines -> tunnel-like channel -> breach (Teton).
- **Uplift pressure** under the dam raised by water in joints (Malpasset).
- **Abutment failure** - wedge failure along intersecting joints / faults; **slope failure** on reservoir rim.
- **Earthquake shaking / fault movement** and **overtopping** due to floods beyond spillway capacity.
- **Slaking / softening** of rock on contact with water (St. Francis conglomerate).

2.5 Indian Dam Examples (geology)

Dam / River	Type	Geological setting
Nagarjuna Sagar (Krishna)	Masonry gravity + earth flanks	Hard Precambrian rocks of Nallamala region
Srisailam (Krishna)	Concrete gravity	Narrow gorge in quartzite / crystalline rocks
Idukki (Periyar)	Double-curved **arch**	Narrow gorge in **massive charnockite**
Tehri (Bhagirathi)	Earth + rock-fill (260 m)	**Seismic zone IV**; phyllite and quartzite of lesser Himalaya
Somasila (Penna)	Earthen	Wide valley, Cuddapah / granite-gneiss terrain
Koyna (Koyna)	Rubble-concrete gravity	**Deccan basalt**; RIS 1967

3. Reservoirs

A **reservoir** is an **artificial lake** formed by storing water behind a dam. **Purposes:** water storage (irrigation, supply), hydropower, flood regulation, recreation.

3.1 Geological Considerations for Successful Reservoirs

- **Water-tightness (no leakage):** reservoir basin should be formed in **impervious rocks** (shale, clay, massive granite, gneiss) or by **impermeable beds**. **Avoid** karst limestone,

highly jointed basalt, fractured quartzite, coarse alluvium on the rim.

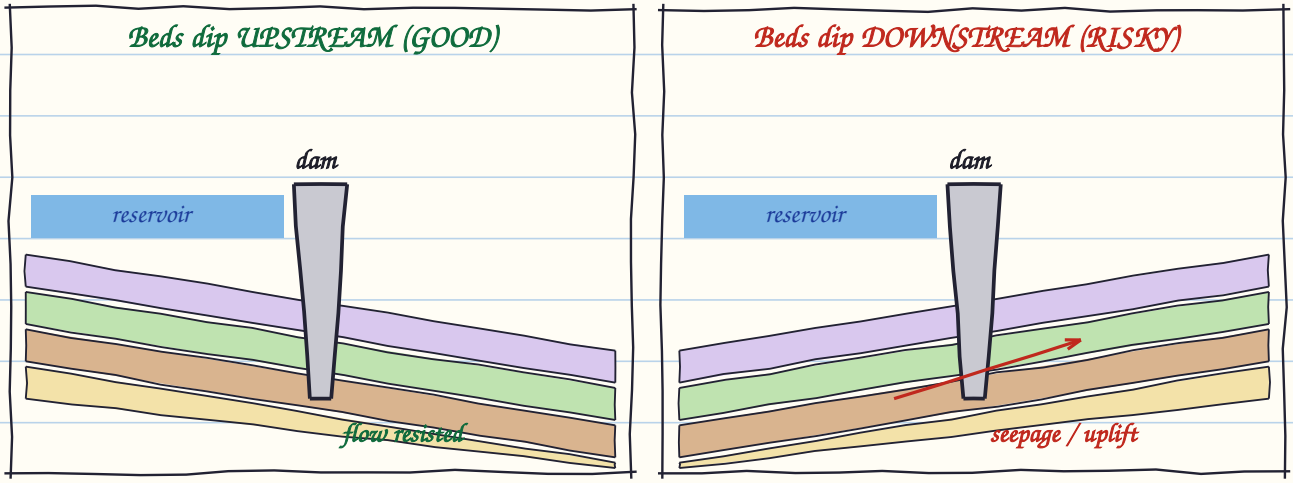


Fig 2: Effect of dip of beds on seepage under the dam

- **Structure of rocks:**
  - **Dip of strata:** beds dipping **upstream / towards the reservoir** and **water-bearing strata dipping away** -> **leakage possible**; beds dipping **downstream** and sloping towards valley -> **water escapes** (risk) if permeable; **impervious beds dipping upstream** are best.
  - **Faults / joints** crossing the rim or the valley can act as leakage channels; **dykes** may be barriers.
  - **Anticline** - leakage through the limbs; **syncline** - water-tight, favourable if impermeable beds.
- **Water table:** if the **ground water table lies above the reservoir level** around the rim -> **no leakage** (water table slopes towards the reservoir). If the water table is **below reservoir level and slopes away** (steep downstream) -> **leakage** (e.g., in basins with deep water table).
- **Stability of reservoir slopes (rim):** unstable rim -> **landslides into the reservoir** (Vajont) -> waves and loss of capacity; study **slope, weathering, strata, ground-water conditions**.
- **Seismicity:** avoid **active fault zones**; **reservoir-induced seismicity (RIS)** (Koyna, Kariba, Hoover) after impounding of **deep reservoirs (> 100 m)** on pre-stressed faults.
- **Siltation:** rock type and land use in **catchment** control silt load; **soft rocks, steep slopes, deforestation** produce silt -> **reduces storage** capacity.
- **Water quality:** soluble rocks (**gypsum, rock salt**) may spoil water chemistry; dissolved

minerals.

- **Large storage capacity:** broad valley behind a narrow dam gorge, with sealed rim.
- **Submergence:** minimise loss of land, forests, mineral resources, historic sites; **rehabilitation**.

3.2 Treatment of Leakage

- **Grouting (curtain)** and **blanketing** (clay blanket on the upstream floor), **cut-off walls**, **sealing of springs**, **concrete lining**, plugging of sinkholes; careful **site selection**.

3.3 Life of Reservoirs

- **Storage capacity** declines because **sediment (silt) enters and settles** in the reservoir - **reservoir sedimentation**. Life = time to fill the **dead storage** (and then live storage) with silt.
- Sedimentation rate depends on: **catchment area, slope, rock type, soil erosion, rainfall, vegetation**, land use, size of reservoir, **trap efficiency** (fine particles escape).
- **Life (years) ≈ Total storage capacity / Average annual silt inflow**. Typical **design life 50-100 years**; Indian reservoirs lose ~ **0.5-1% capacity per year**.
- **Zones of reservoir: dead storage** (below minimum draw-down level), **live storage** (useful), **flood storage** (surcharge).
- **Control of sedimentation: catchment treatment** (afforestation, check dams, contour bunding, terracing), **silt traps / sediment ponds**, **dredging**, **sluicing and flushing**, **bypass of silty floods**, **upstream sediment detention dams**, **optimum operation**.

Examples

- Nagarjuna Sagar (Krishna River, A.P./Telangana) - 124 m high masonry gravity dam founded on hard **Precambrian rocks**; Srisailam - concrete gravity dam in a narrow **gorge through the Nallamala hills**; Tungabhadra - heavy silting has reduced its storage capacity.

3.4 Reservoir-Induced Seismicity (RIS)

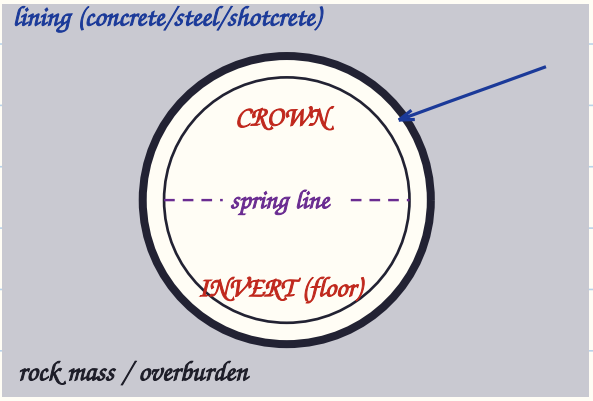
- **Earthquakes triggered by impounding** of a reservoir - pore-pressure rise along existing faults + extra load of water.
- Known cases: **Koyna (India, 1967, M 6.5)**, Kariba (Zambia, 1963), Hoover (USA), Aswan, Hsinfengkiang (China).
- Favourable conditions: **depth of water > 100 m**, large volume, **pre-existing fractures**,

rapid filling.

- Precautions: **seismic study of region**, **install seismographs**, **fill slowly in stages**, flexible dam design.

4. Tunnels

A **tunnel** is an **underground passage (horizontal or nearly so) excavated through rock or soil**, **without removing the overburden**, with a cross-section generally closed.



- Circular - soft rock, water pressure
- Horseshoe - roads / railways
- D-shaped / arched - hard rock
- Rectangular - cut & cover, metro

Shape depends on rock quality, use and construction method

Fig 3: Tunnel cross-section and shapes

4.1 Purpose of Tunnelling

- **Transportation: railway, highway** tunnels through mountains (Konkan Railway, Jawahar tunnel, Rohtang Atal tunnel), **metro** (urban).
- **Water conveyance: irrigation, hydro-power (head race / tail race) tunnels**, **water supply** and **sewage** tunnels (e.g., the **SLBC and Veligonda project tunnels** in the Krishna basin).
- **Mining** (adits, drifts), **military and defence**, **storage** (oil, gas, nuclear waste), **utility tunnels** (cables, pipes), **pedestrian subways**.
- **Advantages:** shorter route, avoid steep gradients, protect from weather / landslide, preserve surface land, save time and fuel.

4.2 Effects of Tunnelling (on rock and environment)

- **Stress redistribution** around the opening -> **roof fall, rock burst, squeezing / swelling** of rock, **over-break**.
- **Ground water: inflow** (flooding), **lowering of the water table** (drying of wells and springs), **subsidence** of surface (urban tunnels), **blasting vibration**.

- **Gases:** methane, H_2S, CO_2 in coal / sedimentary rocks; **heat** in deep tunnels (geothermal gradient, 1° C per 30 m).
- **Environmental:** disposal of **muck**, dust, noise, vibration, ecological impacts.

4.3 Lining of Tunnels

Lining = a **supporting layer** put on the tunnel surface to **prevent collapse, resist rock and water pressure, reduce friction (water tunnels) and give a smooth, durable surface.**

- **Purposes:** support the **roof & walls**, prevent **fall of loose rock**, exclude water (**water-tightness**), give **smooth flow** in conduits, **protect against weathering** (slaking), safe working.
- **Types and materials:**
  - **Timber / steel supports** (arches, ribs, **steel sets**) - temporary support; used in weak ground.
  - **Shotcrete (sprayed concrete)** with **wire mesh / steel fibre** and **rock bolts** - flexible lining used in **NATM** (New Austrian Tunnelling Method).
  - **Cast-in-situ (plain or reinforced) concrete** - permanent lining; **pre-cast concrete segments** - for **TBM** (tunnel boring machine) tunnels, esp. soft ground.
  - **Brick / masonry lining** - old railway tunnels; **steel liner (penstock)** - high-pressure hydropower; **cast iron segments** - metro.
- **Thickness:** from **thin (shotcrete 5-15 cm)** to **thick (concrete 30-100 cm)** depending on **ground conditions and tunnel size.**
- **Need of lining depends on geology: hard, massive rock (granite, quartzite) - unlined or partially lined; weak, jointed, weathered, water-bearing rock - full lining** essential.

Rock condition	Support needed	Example
Massive hard rock	None / spot bolts	Granite, quartzite
Fractured rock	Bolts + shotcrete + mesh	Jointed basalt
Weak / squeezing rock	Steel ribs + concrete lining	Shale, schist, fault zone
Soft ground	Segmental lining (TBM)	Alluvium, clay (metro)

4.4 Influence of Geology on Successful Tunnelling

1. **Rock type & strength: strong, massive, homogeneous** rocks (granite, gneiss, quartzite, dolerite, massive limestone, sandstone) are **ideal** - stable, self-supporting,

need little support. **Weak rocks** (shale, schist, slate, phyllite, clay, weathered rock) need heavy support.

- **Soft ground** (soil, alluvium) - **shield / TBM**; **soluble rocks** (limestone, gypsum) - caves.
- **Rock hardness affects drilling** and blasting cost; **abrasiveness** wears cutters; **rippability** (seismic velocity).

2. **Geological structures**
   - **Attitude of beds: horizontal beds** - roof stays good if strong beds above; **dipping beds** - best when the tunnel axis is **perpendicular to strike (across the strata)**; tunnel **along the strike** is risky: **roof and one wall slide** along bedding planes and **uneven pressure** develops.
   - **Folds:** in an **anticline**: tension joints, **roof falls**; in a **syncline**: good rock but **water accumulation**; best to **cross folds at right angles** and in the **limb zone** with moderate dip. Tunnel along the axis of a **plunging fold** can be good if beds are strong.
   - **Faults: major problem** - crushed rock (gouge), **water inflow**, **heavy support**, squeezing; **cross faults at high angle** and plan treatment (**probe holes, grouting, spiling**); avoid **parallel** alignment with a fault.
   - **Joints: joint orientation** affects wedge falls; **closely spaced** - over-break; tunnel **across** joint sets; **grouting** if water bearing.
   - **Unconformities & dykes:** variable ground; **dykes** may supply or block water.
3. **Ground water: inflow is the most serious problem** - flooding, **slow progress**, instability, **lining pressure**; **water-bearing joints, faults, sand layers, karst**; control by **grouting, dewatering, drainage, freezing**; tunnels below water-table need **water-tight lining**.
4. **Weathering and overburden:** deep, **weathered** zone gives poor roof; **thickness of cover** (at least **1.5-2 × tunnel diameter**) for stability; **portals** should be located on **sound rock**, away from landslide / debris slopes.
5. **Rock stress and temperature: high cover -> rock burst** (hard rock) or **squeezing** (soft rock); heat; **seismic activity**.
6. **Gases & special rocks: methane** in coal measures, **swelling clays (montmorillonite)**

soluble minerals, radon.

7. **Rock mass quality:** assessed by **RQD, RMR, Q-system** - used to decide **support**; **pre-construction exploration** (bore-holes, **seismic refraction**, resistivity) for **geological longitudinal section**.
8. **Construction method choice: drill & blast** (hard rock), **NATM, TBM** (uniform rock), **cut & cover** (shallow urban), **shield tunnelling** (soft ground).

Alignment rules (remember!)

- (i) Cross rock structures at **high angle**. (ii) Avoid **faults, shear zones & weathered ground**. (iii) Keep **portals in sound rock**. (iv) Stay above/below **aquifers** as needed. (v) Prefer **strong, homogeneous** rock.

4.5 Geological Investigation for Tunnels

- **Geological mapping** of the surface along the alignment (rock types, dip, strike, structures, springs), **aerial photo / remote sensing**.
- **Geophysical surveys** (**resistivity, seismic refraction**) for **overburden depth, faults, water**.
- **Exploratory drilling** and **core logging (RQD)**; **pilot tunnels**, **probe holes ahead of face**.
- Prepare **geological longitudinal section** and **rock mass classification** to forecast **support & cost**.
- **Case: Konkan Railway** (~ 90 tunnels) - geology-based alignment reduced collapse; **Atal Tunnel (Rohtang)** - 9 km highway tunnel built through **fractured, water-bearing Himalayan rock and the Seri Nala fault zone** with heavy support.

4.6 Rock Mass Classification for Tunnels

RMR value	Class	Support (about 10 m dia. tunnel)
81-100	I - Very good	None / spot bolts
61-80	II - Good	Local bolts, thin shotcrete
41-60	III - Fair	Systematic bolts, 5-10 cm shotcrete
21-40	IV - Poor	Bolts + 10-15 cm shotcrete + steel ribs
< 20	V - Very poor	Heavy ribs, thick lining, pre-support

- **RMR (Bieniawski)** uses 5 parameters: strength of intact rock, **RQD**, **joint spacing**, **joint condition**, **ground water**, + orientation adjustment. **Q-system (Barton)** $Q = (RQD/Jn) \times (Jr/Ja) \times (Jw/SRF)$.

4.7 Tunnel Hazards and Remedies

Hazard	Geological cause	Remedy
Roof fall / over-break	Joints, faults, weathered rock	Rock bolts, shotcrete, steel ribs
Squeezing	Soft shale / schist under high cover	Yielding supports, heavy lining
Water inflow	Fractures, karst, faults, sand	Grouting, drains, dewatering
Rock burst	Brittle rock, high stress	Destressing, support, slow advance
Gas / heat	Coal measures, deep cover	Ventilation, monitoring
Swelling	Clays (montmorillonite)	Invert lining, drainage

5. Comparison and Summary

Aspect	Dam	Reservoir	Tunnel
Key rock need	Strong, sound foundation	Impervious basin	Strong, stable, dry rock
Main structural risk	Faults, downstream dip	Leaky joints, karst	Faults, joints along axis
Water issue	Seepage and uplift	Leakage / siltation	Inflow
Treatment	Grouting, drains	Blanketing, grouting	Lining, drainage, bolts
Failure case	St. Francis, Malpasset	Vajont (rim slide)	Squeezing & collapse

5.1 Short Revision Questions

1. **Name the purposes of a dam.** Irrigation, power, water supply, flood control, navigation, recreation.
2. **Which dam needs the strongest foundation?** Arch and gravity (concrete / masonry) dams; arch dams need strong abutments.
3. **Why is upstream dip preferred?** Water pressure pushes beds into the rock, seepage is resisted and uplift is reduced.
4. **What is the life of a reservoir?** Time taken for sediment to fill the reservoir (storage / annual silt inflow).
5. **How do we reduce reservoir leakage?** Grouting, clay blanket, cut-off wall, and selecting impervious rim.

6. **Purpose of tunnel lining?** Support, water-tightness, smooth flow and protection.
7. **Which attitude of beds is best for tunnelling?** Tunnel axis across (perpendicular to) the strike in strong, dry beds.

Final exam tips

- For Unit V always give: (1) a neat sketch, (2) geological factors in bullet form (rock type - structure - water - weathering - seismicity), (3) one Indian example, (4) remedial measure (grouting / lining).